1. Experiment in short

Number of nodes	4
Number of topologies	2
Location of the experiment	Hadera coal pier, Israel $(32.47251°N, 34.86391°E)$
Measurement duration	45 minutes
Measurement granularity	1 second
Link quality measurements	Received Signal Strength Indicator

2. Location

The experiment took place on May 5, 2017, in the proximity of the Hadera coal pier, in Israel. The water depth in the experiment are was about 25 m. A Google map illustration of the experimental setup is shown in Fig. 1. The experiment included 4 nodes, communicating using three different types of EvoLogics modems. The nodes were deployed from a rubber boat (Node 1), from a working boat (Node 3), and from the pier (Node 2 and 4). Nodes 2 and 4 were placed 150 m apart at a depth of 5 m. By placing the working boat (Node 3) at different locations, we changed the network topology.

During the experiments, all the propellers, engines and echo-sounders were turned off, leaving both the working boat and the rubber boat drifting.

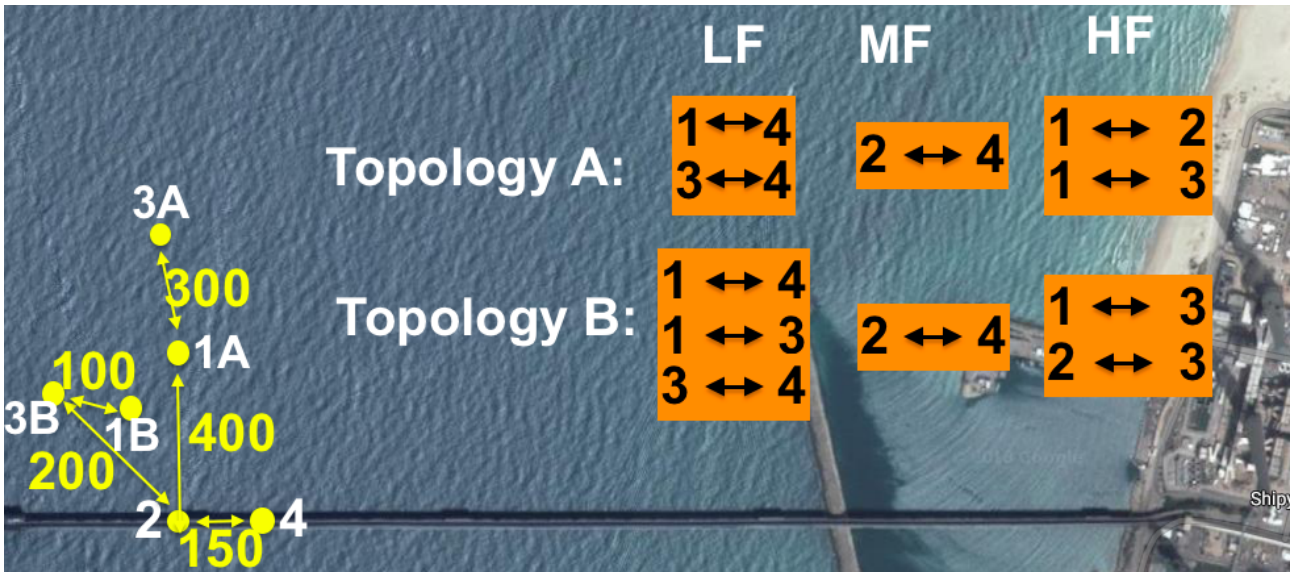


Fig. 1: Satellite picture of the sea trial location and logical view of the topology.

3. TOPOLOGIES

The network included four nodes, which shifted from Topology 1 to Topology 2 during the experiment. The network topology is fully connected. In the following, we list the nodes' locations with respect to the positions marked in Fig. 1.

- Node 1 (A) 32.47417 N, 34.86684 E; (B) 32.47249 N, 34.86599 E;
- Node 2 32.47059 N, 34.8668 E;
- Node 3 (A) 32.47648 N, 34.86573 E; (B) 32.47274 N, 34.86409 E
- Node 4 32.47059 N, 34.8684 E

4. DESCRIPTION OF THE NODES

The experiment included 4 nodes, communicating using three different types of EvoLogics modems: the S2C 7-17 modem (low-frequency, LF); S2C 18-34 modem (medium-frequency, MF); the S2C 48-78 modem (high-frequency, HF). A total of 8 modems was deployed: 3 LF, 2 MF and 3 HF. All modems employ EvoLogics's proprietary sweep-spread carrier modulation scheme. For further information, the interested reader is referred to EvoLogics's web site: https://www.evologics.de/en/products/acoustics/index.html. The modems were integrated into the nodes in as shown in 1.

5. TESTED PROTOCOLS AND SCHEMES

The deployment served to test an optimal multimodal TDMA-based schedule that exploited the knowledge of neighboring links in underwater acoustic networks in order to optimally allocate the time-slots of the scheduling protocol. In addition, an Aloha-multimodal MAC protocol was tested as a benchmark scheme.

6. LINK MEASURES

The network links are evaluated by means of the received signal strength indicator (RSSI). Transmissions were carried out using the EvoLogics modems' instant message mode, which makes it possible to send data by excluding any further control exerted by the modem's DMAC protocol. Upon transmission, the RSSI was returned directly by the receiving modems' firmware.

7. DATA AVAILABLE

The stored data is given in the *HaderaMultimodalSched-05-17.mat* file. This file brings together 2424 measurement epochs, which have been taken at constant intervals of 1 second starting from the beginning of each experiment. The measured data is provided in file `BerlinMultimodal-06-16.mat`, which includes all topologies presented in Fig. 1. For each topology, we provide about 600 measurement epochs, given at

constant intervals of 1 second starting from the beginning of each experiment. The file contains four types of matrices:

- the location matrices `LocMat1` and `LocMat2`, of size $1212\times4\times3$;
- the topology measurement matrices `TopMat1` and `TopMat2`, of size $1212\times4\times4\times3$;
- the adjacency matrices `AdjMat1` and `AdjMat2` of size $1212\times4\times4$;
- the technology matrices `TechMat1` and `TechMat2` of size $1212\times4\times3$.

Additionally, for each matrix type, we concatenate all matrices into a single matrix, respectively named:

- `FullLocMat` of size $2424\times4\times3$;
- `FullTopMat` of size $2424\times4\times4\times3$;
- `FullAdjMat` of size $2424\times4\times4$;
- `FullTechMat` of size $2424\times4\times3$.

Using Matlab/Octave matrix notation:

- `LocMat1(t,i,:)` and `LocMat2(t,i,:)` or `FullLocMat(t,i,:)` are the UTM coordinates (zone 36E) and the depth of node i at time instant t;
- `TopMat1(t,i,j,p)` and `TopMat2(t,i,j,p)` or `FullTopMat(t,i,j,p)` is the measured RSSI for technology p over the link $i \rightarrow j$ at time t;
- `AdjMat1(t,i,j)` and `AdjMat2(t,i,j)` or `FullAdjMat(t,i,j)` is equal to 1 if there exists a topology link between nodes i and j at time t;
- `TechMat1(t,i,p)` and `TechMat2(t,i,p)` or `FullTech(t,i,p)` is equal to 1 if node i can use physical layer technology p at time t.

8. Reference to detailed description

Further details about this experiment are given in the following papers:

- F. Campagnaro, P. Casari, M. Zorzi, R. Diamant, "Optimal Transmission Scheduling in Small Multimodal Underwater Networks," in *IEEE Wireless Communications Letters*, vol. 8, no. 2, pp. 2162–2337. Apr. 2019.